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RESEARCH

Catch before they fall: a pose-guided attention framework for indoor safety

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Abstract
Falls can pose a serious health threat, especially for older people, often leading to fractures, head injuries, or long-term disability. This highlights the need for reliable and non-invasive detection systems. Existing solutions often suffer from limitations such as user discomfort with wearables, constrained coverage of fixed sensors, or environmental challenges in vision-based methods. This paper proposes a dual-stream attention-guided robust indoor fall detection framework to solve the problem. The approach combines a pose-guided stream with a video stream, in which the former captures high-level skeletal features through MediaPipe and simultaneously processes spatiotemporal dynamics from raw RGB frames using a 3D Convolutional Neural Network (CNN) with a temporal attention module. In order to enhance classification precision, features from each stream are adaptively fused in a block based on attention mechanisms, which improves the model's interpretation of posture and movement semantics. Testing on the KFall dataset reveals that the proposed method achieves 98.71% accuracy, surpassing pre-existing benchmarks. Triggering a visual alert by displaying a red rectangle on the screen upon the occurrence of this event is a subsequent outcome of this work. An ablation study highlights the effectiveness of each component. Finally, the proposed work advances fall detection by combining pose estimation and attention-based deep learning to deliver an accurate, interpretable, and deployable solution.

Keywords 3DCNN · Attention mechanism · MediaPipe · Pose estimation · Indoor fall detection · Kfall dataset

1 Introduction

Falls represent one of the most critical health hazards across age demographics, with particularly severe implications for older adults. According to the World Health Organization [1], falls are the second leading cause of accidental injury deaths globally, resulting in approximately 684,000 fatalities annually. Among individuals aged 60 and above, the risk of morbidity and long-term disability due to falls is disproportionately high, accounting for a significant burden on healthcare systems and caregivers. The escalating prevalence of falls has necessitated the development of reliable, real-time fall detection systems to enable timely intervention and mitigate injury severity.

Despite significant advancements in health monitoring, existing fall detection systems encounter several challenges [2, 3]. Wearable sensor-based solutions, though popular, suffer from user non-compliance, battery dependency, and discomfort. Stationary sensors like pressure mats and ambient devices have limited coverage and require fixed installation locations. Furthermore, traditional vision-based approaches often struggle with occlusions, low-light conditions, and high computational demands. These limitations hinder the deployment of robust and non-intrusive systems capable of delivering accurate fall detection in indoor environments.

Although prior research [4–7] has explored both wearable and vision-based fall detection techniques, most approaches

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either neglect the temporal dynamics of motion or fail to integrate human skeletal movement meaningfully. Moreover, limited attention has been paid to attention mechanisms that dynamically weigh relevant motion features over time, which is crucial for distinguishing fall events from normal activities like sitting or bending. A notable gap exists in harnessing the synergy of pose estimation and temporal attention within 3D convolutional neural network architectures for precise fall recognition.

The motivation for this study arises from the pressing need for a non-invasive, accurate, and efficient fall detection solution that can be deployed in indoor surveillance settings, especially in smart homes and healthcare facilities. Advances in pose estimation technologies and attention-based deep learning models provide an opportunity to extract detailed joint and motion information while focusing on the most salient temporal features. This motivates the design of a hybrid framework that integrates human pose data with spatio-temporal attention for robust fall recognition.

To address the shortcomings of existing fall detection systems, this work proposes a dual-stream attention-guided framework that leverages both skeletal pose information and spatiotemporal features extracted from raw videos for robust and accurate fall recognition. The major contributions of this study are as follows:

- **Dual-Stream Architecture:** A novel framework is introduced that employs two parallel processing streams:
  - The pose-guided stream captures postural dynamics by extracting high-level skeletal features such as joint coordinates, angular displacements, inter-joint distances, motion trajectories, and acceleration using MediaPipe.
  - Spatio-temporal dynamics are stored in video streams as RGB raw video frames and processed with a 3D Convolutional Neural Network (3D CNN) with temporal attention for motion pattern recognition over time.
- **Cross-Stream Feature Fusion:** The model integrates both stream features with attention-based feature integration blocks, enabling the joint learning of context related to postures and movement semantics for more reliable classification.
- **Experimental Validation and Analysis:** In-depth testing has been carried out on the KFall dataset, proving the model's superiority against previous benchmarks. An accuracy of 98.71% is achieved, validating the effectiveness of the hybrid approach.
- **Ablation Study:** A comprehensive ablation study explores the impact of each module on performance, supporting the proposed mechanism's effectiveness.

The rest of the paper is structured as follows: Sect. 2 includes related work and highlights current limitations in fall detection methodologies. Section 3 describes the proposed methodology, including data preprocessing, pose estimation, model architecture, feature fusion strategy, and classification. Section 4 explores the experimental setup, results, ablation study, and performance comparisons. Finally, Sect. 5 concludes the paper with key findings and future research directions.

2 Related work

The health concern of falls poses a risk to the older adult population, and ample systems have been developed for detection. The evolution of fall detection technology has made a journey from basic wearable sensors to sophisticated computer vision and deep learning techniques. Wearable healthcare devices solve intrusiveness, but their accuracy is often contingent on external factors such as sensor quality, positioning, and user adherence. Because of this, there is still a need to test these systems to evaluate their functionality and user-friendliness practically. Acceptance of fall detection technologies is impacted by older adults by the simplicity, responsiveness, and level of privacy permeated into the system.

The existing literature offers several solutions to the problem, each presenting unique insights and approaches.

Yu et al. [4] developed a fall detection system using Dynamic Vision Sensors (DVS), which significantly alleviates the privacy concerns of camera-based systems. The authors used Spiking Neural Networks (SNNs) to process the event streams and proposed a model with SNNs and 3D-CNNs called NeuCube, achieving accuracy rates of 94.59% and 97.24%, respectively.

Kulurkar et al. [5] did real-time fall classification using low-cost wearable sensors from MbientLab, Apache Flink, and an LSTM network. Their model was based on the public MobiAct dataset and focused on optimizing sensor placement, sampling rates, and achieving effective fall classification. Open-source tools and hardware proved useful in this area as they achieved 95.87

A robotic system capable of environment mapping and fall detection was constructed by Elwaly et al. [8] by integrating 2D LiDAR, IMUs, and cameras into a multimodal sensor suite preset. Their design incorporated advanced path planning methods such as Hector SLAM and RRT. Using YOLOv7, the system was able to achieve fall detection with an accuracy of 96%, while localization accuracy was maintained at 93.8%, proving reliable within dynamic indoor settings.

Qu et al. [9] designed the fall detection system TSFAllDetect, configured with an embedded sensor module, a mobile

deep learning unit, and a server data management system. They proposed a new model for deep learning that outperformed classification neural networks using time-series data from inertial and pressure sensors and achieved better results than traditional sequential classifiers. Their public and custom dataset experiments showed the system to be efficient and versatile in real-world scenarios.

Guo et al. [10] proposed PiFall model, which integrates a ResNet(2+1)D model and pressure-sensitive insoles. They converted data from the insole sensors to time-series stress imaging, which was suitable for processing by CNN. Their study involved 12 distinct activities across five categories, including fall-like events designed to test system sensitivity. PiFall achieved a 91% overall accuracy in fall recognition and 94% accuracy in identifying specific fall types. Moreover, a structured user survey with older adults provided insights into comfort and user satisfaction, emphasizing the importance of human-centred design.

Wang et al. [6] presented a fall detection method based on ultra-wideband (UWB) radar. Their system applied an adaptive channel selection algorithm to isolate motion signals from noise. Time-frequency representations of these signals were processed by a lightweight CNN capable of identifying three distinct fall scenarios (standing, bowing, squatting to fall) with an accuracy of 95.7%.

Finally, Mahmoud et al. [11] proposed a wearable-based fall prediction and detection framework using IMUs with accelerometers, gyroscopes, and magnetometers. Trained on the UPFALL dataset, their system supports proactive fall warnings and real-time alerting via a connected monitoring center. Multiple machine learning algorithms, including K-Nearest Neighbors (KNN), SVM, Random Forest, and Gradient Boosting, were evaluated, with KNN achieving the highest accuracy at 93.5%.

Besides fall detection, human activity recognition (HAR) research also addresses abnormal activity detection, where falls may be considered a critical category of abnormal events. Several HAR studies have contributed to this field by developing advanced models for detecting unusual activities. For instance, Kurchaniya and Kumar [12] proposed a two-stream deep neural network that processes spatial and temporal features separately, using an effective fusion strategy to improve classification accuracy in abnormal activity detection. Ansari and Singh [13] noted that model convergence and performance were enhanced by tuned parameters in recurrent learning models for classifying shoplifting activities. Also, Ansari et al. [3] presented a new architecture for shoplifting detection in megastores, which they named PosturalNet, specifically designed to capture and track accurate postures associated with complex environments. These works highlight the increasing relevance of HAR-based architectures, which can be tailored for improved fall detection. Phyo et al. [7] developed a deep learning model for human activity recognition using skeletal joint motion. Their approach captures spatial-temporal features effectively, improving classification accuracy. The method shows strong potential for use in consumer electronics and real-time applications. Ahn et al. [14] propose the Star-Transformer, a novel architecture that integrates spatio-temporal cross-attention to enhance human action recognition. The model efficiently captures global contextual dependencies while maintaining computational efficiency. Experimental results demonstrate state-of-the-art performance on several benchmark datasets, highlighting its effectiveness in complex action understanding tasks.

Prior research has extensively explored fall detection using wearable sensors, vision-based methods, and deep learning techniques in this study area. Many existing approaches, however, struggle with challenges such as occlusions, varying lighting conditions, and limited temporal modeling, which reduce their accuracy and robustness in complex indoor environments. Studies have applied 3D CNNs or pose estimation separately, but very few integrate these techniques with temporal attention mechanisms that dynamically focus on critical motion cues. Our proposed model addresses these gaps by combining a 3D CNN with a temporal attention module and pose-guided feature extraction. This enables simultaneous analysis of spatial-temporal patterns and detailed skeletal movements. This hybrid framework enhances the detection of fall events with greater reliability and robustness. Moreover, its non-intrusive design supports real-world indoor deployment, providing a practical and advanced solution for elderly fall detection.

3 Proposed methodology

The proposed fall detection framework synergistically combines pose estimation with deep learning enhanced by an attentional mechanism to achieve accurate, robust, and real-time fall detection. The overall system architecture is depicted in Fig. 1. The pipeline begins with acquiring video data from the KFall dataset, which serves as the foundational dataset for the model's training and performance evaluation.

3.1 Overview of the framework

This article proposed a dual-stream network to identify falls, where one stream encodes postural information, and the second encodes spatial and temporal features. Following this approach, our methodology begins with comprehensive preprocessing of raw video sequences to enhance frame quality and standardize input dimensions. The framework is divided into two parallel branches developed to capture distinct but complementary feature representations. The first branch employs the MediaPipe framework to collect 33 key

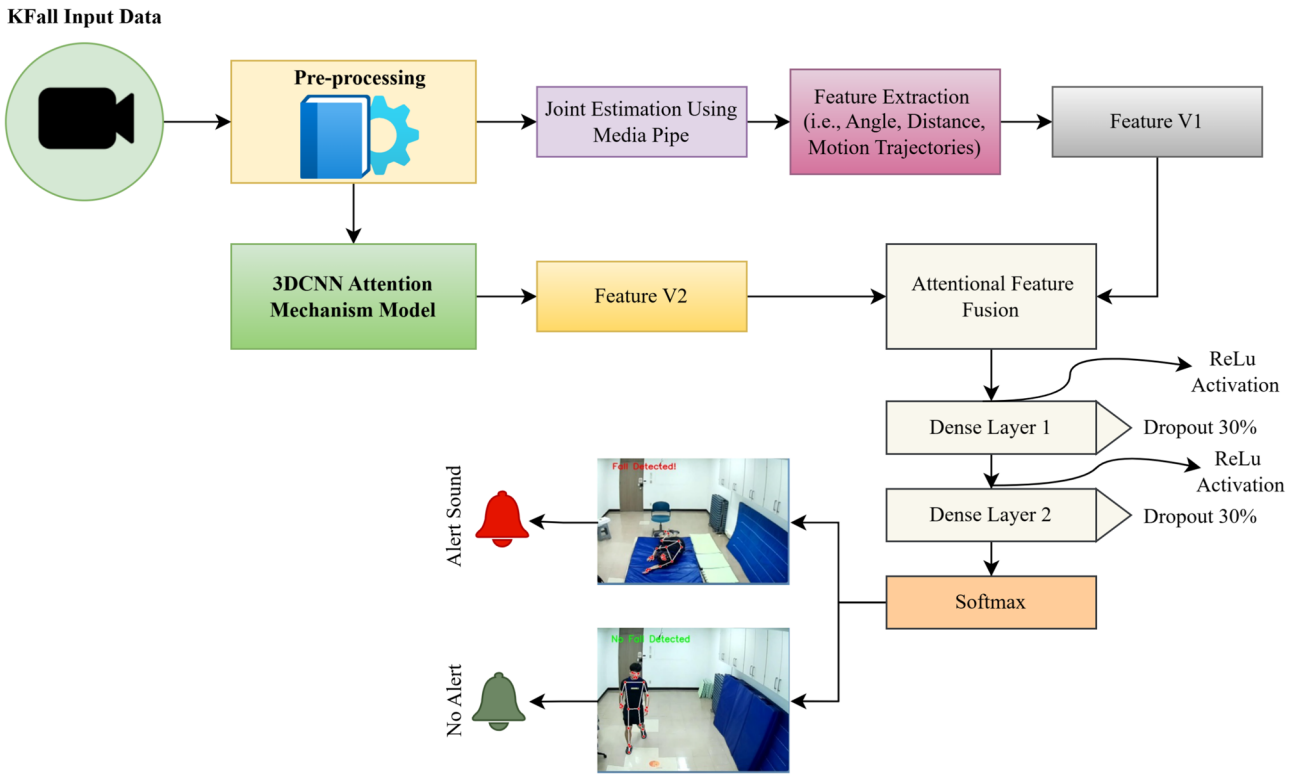


Fig. 1 Proposed 3DCNN attention model for fall detection with pose estimation

2D human pose landmarks corresponding to each video frame. Motion features are engineered from these landmarks, including joint angles, inter-joint distances, velocity vectors, and motion trajectories over time, dubbed as Feature Set V1. This feature set captures skeletal movement dynamics pertinent to fall detection. Concurrently, the second branch applies a 3D Convolutional Neural Network (3DCNN) with a temporal attention mechanism to the preprocessed video frames. This 3DCNN learns spatiotemporal patterns through convolution across spatial and temporal dimensions. At the same time, the attention module dynamically weights frames relative to their importance for fall events, creating Feature Set V2 from pixel data. The created feature sets V1 and V2 are fused by an attentional feature fusion module that autonomously combines the engineered pose-based features and deep-learned spatiotemporal representations into a single embedding. The resulting fused vector undergoes processing through fully connected dense layers with ReLU activations and 30% dropout regularization to enhance generalization and mitigate overfitting. A final Softmax layer determines if there is a fall or none in the input information. When a fall is detected, an alert sound is triggered to notify caregivers, while annotated video frames displaying the detection result are output for interpretability and real-time monitoring. This hybrid architecture, combining explicit pose analytics with an attention-guided 3DCNN, selectively focuses on critical temporal segments, enhancing robustness and accuracy for practical indoor fall detection deployments in elderly care facilities and hospitals. The technicalities used in this methodology are discussed here in brief.

3.2 Input: raw video sequences

The system takes raw video sequences capturing human activity as input. These videos often vary in length, resolution, and frame rate. To ensure uniformity and improve downstream processing, the raw videos undergo preprocessing.

3.3 Data preprocessing

All video frames are resized to a fixed dimension of 224×224 pixels and normalized to a pixel value range of 0 to 1, ensuring consistent input quality for the model. To handle variations in video length and improve temporal uniformity, a frame-based method is applied whereby each video is sampled or tested to contain exactly 100 frames. This approach standardizes both spatial and temporal aspects of the input data, enhancing the robustness and reliability of the fall detection framework during training and evaluation.

3.4 Dual-stream feature extraction

After preprocessing, the framework splits into two parallel streams, each specializing in different feature representations. The first stream here encodes posture information, while the other one deals with the spatial-temporal dynamics of any action. A detailed description of both streams is presented below.

3.4.1 Pose estimation and feature engineering

This stream extracts explicit pose-encoded features from the human skeleton through video frames. Here, we first extract skeletal information from video frames for humans and then encode the features. The techniques used to perform this task are discussed below.

Pose estimation. The MediaPipe framework [15] is utilized for real-time 2D human pose estimation. MediaPipe employs a multi-stage pipeline combining a BlazePose model with a lightweight convolutional neural network (CNN) to detect and track 33 body landmarks, including key joints such as shoulders, elbows, wrists, hips, knees, and ankles. Each landmark is represented as a coordinate vector:

$$\mathbf{p}_i = (x_i, y_i), \quad i = 1, 2, \ldots, 33 \tag{1}$$

where x_i, y_i are the normalized pixel coordinates of the ith landmark in the image frame.

The MediaPipe pipeline for pose estimation consists of three main components. First, the Detection Module identifies the presence of a human body within each frame and provides an approximate bounding box to localize the person. Using an employed regression-based CNN strategy trained on annotated datasets, the Landmark Model accurately predicts the x and y coordinates of 33 specific joints, including the shoulders, wrists, and other anatomical markers. The Tracking and Smoothing module applies temporal filtering and tracking heuristics to smooth and stabilize the landmark positions to reduce noise and jitter as much as possible. This framework enables precise capture and long-term evaluation of the motion and posture of limbs over substantial time durations. The consequence of MediaPipe for pose estimation on processing the input image is depicted in Fig. 2, where the silhouette of an individual has all 33 landmarks prominently placed and marked on it. Furthermore, the figure showcases MediaPipe's functionality on actual images, proving its capability to detect and track joints in real-life situations consistently.

Feature engineering. From the extracted pose landmarks, several handcrafted features are computed to encode spatial and temporal aspects of body motion:

- **Joint Angles:** The angle θ_{ijk} formed at joint j by adjacent joints i and k is calculated as:

$$\theta_{ijk} = \cos^{-1}\left(\frac{(\mathbf{p}_i - \mathbf{p}_j)\cdot(\mathbf{p}_k - \mathbf{p}_j)}{\|\mathbf{p}_i - \mathbf{p}_j\| \cdot \|\mathbf{p}_k - \mathbf{p}_j\|}\right) \tag{2}$$

  This captures the relative posture and bending at key joints such as knees and elbows.
- **Inter-Joint Distances:** Euclidean distances d_{ij} between pairs of landmarks i and j provide information on body configuration and expansion:

$$d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\| = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{3}$$

- **Velocity Vectors:** The temporal change in joint positions captures motion dynamics. For landmark i at frame t, velocity $\mathbf{v}_i^{(t)}$ is approximated as:

$$\mathbf{v}_i^{(t)} = \mathbf{p}_i^{(t)} - \mathbf{p}_i^{(t-1)} \tag{4}$$

  reflecting speed and direction of joint movement.
- **Motion Trajectories and Acceleration:** Sequential positions of each landmark across a temporal window T form a trajectory:

$$\mathcal{T}_i = \{\mathbf{p}_i^{(t)}, \mathbf{p}_i^{(t+1)}, \ldots, \mathbf{p}_i^{(t+T-1)}\} \tag{5}$$

  which encodes the path traced by joints over time. To capture the second-order dynamics of motion, acceleration is also computed as the difference of successive velocities:

$$\mathbf{a}_i^{(t)} = \mathbf{v}_i^{(t)} - \mathbf{v}_i^{(t-1)} = \mathbf{p}_i^{(t)} - 2\mathbf{p}_i^{(t-1)} + \mathbf{p}_i^{(t-2)} \tag{6}$$

  Such features are useful for noticing more impulsive, sudden movements like quick downward accelerations during falls where position or velocity data becomes less informative.

All these features are concatenated to form the handcrafted feature vector $\mathbf{F}_{V1}$ for each video segment:

$$\mathbf{F}_{V1} = [\theta, d, \mathbf{v}, \mathcal{T}, \mathbf{a}] \tag{7}$$

where θ, d, $\mathbf{v}$, $\mathcal{T}$, and $\mathbf{a}$ represent the sets of joint angles, distances, velocities, trajectories, and accelerations, respectively.

The process of evaluating and encoding postural features from video frames is also systematically detailed in Algorithm 1. This algorithm captures key skeletal dynamics by extracting joint angles, inter-joint distances, velocities, trajectories, and accelerations, which collectively form a comprehensive handcrafted feature vector for fall detection. The explicit representation of skeletal motion through

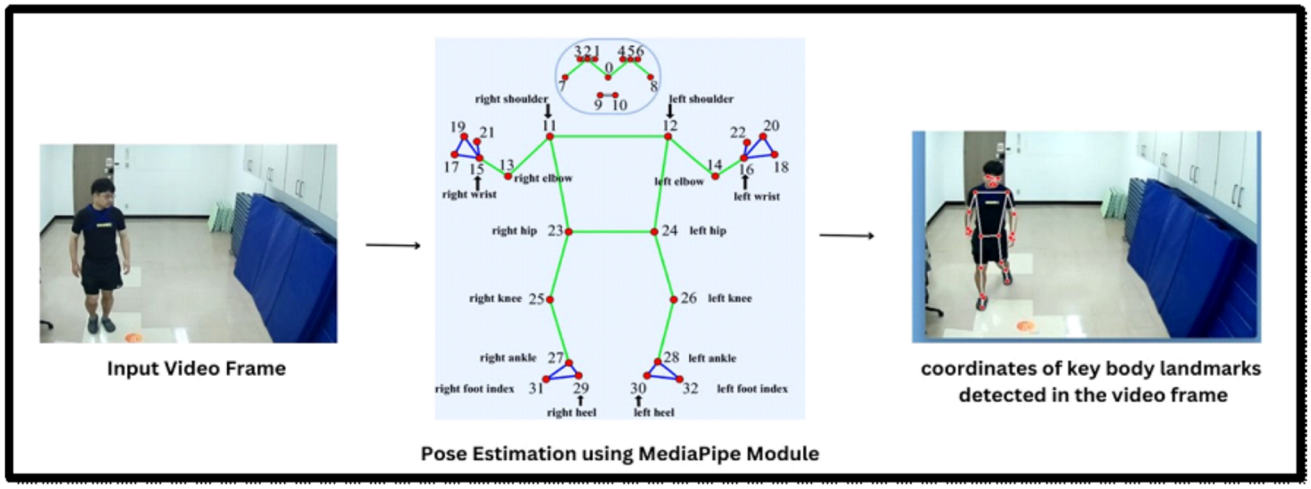


Fig. 2 Pose estimation using MediaPipe

these pose-based engineered features enables the model to focus on meaningful biomechanical changes characteristic of fall events, providing complementary information to deep-learned visual features.

Algorithm 1 Pose Estimation and Feature Engineering

Input: Video frames $\{I_t\}$ for $t = 1$ to T, Where T=100
Output: Handcrafted feature vector $\mathbf{F}_{V1}$

for $t = 1$ to T **do**
  $\mathbf{p}_i^{(t)} \leftarrow$ MediaPipePoseEstimation(I_t) ▷ Detect 33 body landmarks
for $t = 1$ to T **do**
  for each joint triplet (i, j, k) **do**
    $\theta_{ijk}^{(t)} \leftarrow$ computeJointAngle$(\mathbf{p}_i^{(t)}, \mathbf{p}_j^{(t)}, \mathbf{p}_k^{(t)})$
  for each landmark pair (i, j) **do**
    $d_{ij}^{(t)} \leftarrow$ EuclideanDistance$(\mathbf{p}_i^{(t)}, \mathbf{p}_j^{(t)})$
  if $t > 1$ **then**
    for each landmark i **do**
      $\mathbf{v}_i^{(t)} \leftarrow \mathbf{p}_i^{(t)} - \mathbf{p}_i^{(t-1)}$ ▷ Velocity vector
  if $t > 2$ **then**
    for each landmark i **do**
      $\mathbf{a}_i^{(t)} \leftarrow \mathbf{v}_i^{(t)} - \mathbf{v}_i^{(t-1)}$ ▷ Acceleration vector
for each landmark i **do**
  $\mathcal{T}_i \leftarrow \{\mathbf{p}_i^{(1)}, \mathbf{p}_i^{(2)}, \ldots, \mathbf{p}_i^{(T)}\}$ ▷ Trajectory over time window T
$\mathbf{F}_{V1} \leftarrow$ concatenate$(\theta, d, \mathbf{v}, \mathcal{T}, \mathbf{a})$ ▷ Combine all features
return $\mathbf{F}_{V1}$

3.4.2 Spatio-temporal feature learning via 3D CNN with temporal attention

To robustly capture the spatial and temporal features of action dynamics for any person, especially those associated with fall events, we integrate a 3D convolutional neural network (3DCNN) with a temporal attention mechanism, as shown in Fig. 3. This module processes raw video sequences to extract compact, informative feature representations that are highly discriminative for fall detection.

3DCNN backbone The pipeline begins by feeding the input video sequence into a stack of 3D convolutional layers, which apply convolutional filters across both spatial and temporal dimensions. Given an input tensor $X \in \mathbb{R}^{T\times H\times W\times C}$, where T is the number of frames, $H \times W$ is the spatial resolution, and C denotes the number of channels, the 3D convolutional operation at the lth layer is defined as:

$$Z^{[l]} = f(W^{[l]} * X^{[l-1]} + b^{[l]}) \tag{8}$$

where $*$ denotes 3D convolution, $f(\cdot)$ is a non-linear activation function (e.g., ReLU), and $W^{[l]}$, $b^{[l]}$ are the learnable weights and biases of the layer.

To reduce the spatial and temporal dimensions and improve generalization, each convolutional layer is followed by a 3D max-pooling operation:

$$A^{[l]} = \text{MaxPooling}(Z^{[l]}) \tag{9}$$

These layers progressively abstract spatiotemporal patterns from the input sequence, producing a condensed feature map across frames.

Temporal attention block Following the final pooling operation, the intermediate feature tensor is passed through a *temporal attention block*, which selectively emphasizes temporally informative frames that contribute most strongly to recognizing fall events. This block has three main components: *feature extraction*, *attention layer*, and *temporal integration*.

- **Feature Extraction:** Let the output of the final pooling layer be $A \in \mathbb{R}^{T\times H\times W\times D}$. Each frame's feature map A_t

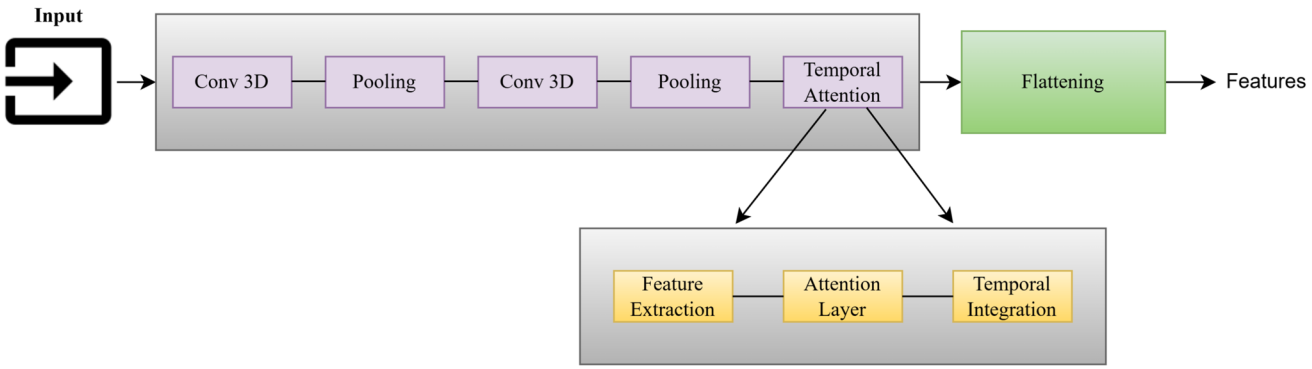


Fig. 3 Attention layer representation in 3DCNN

is flattened to obtain:

$$F_t = \text{Flatten}(A_t), \quad \text{for } t = 1, 2, \ldots, T \tag{10}$$

resulting in a matrix $F \in \mathbb{R}^{T \times (HWD)}$, where each row F_t represents the feature vector of a single frame.

- **Attention Layer:** The attention weights α_t are computed using a feed-forward attention mechanism:

$$e_t = \mathbf{w}^\top \cdot \tanh(\mathbf{W} \cdot F_t + \mathbf{b}) \tag{11}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{12}$$

where $\mathbf{W} \in \mathbb{R}^{d_a \times HWD}$ is a learned projection matrix, $\mathbf{b} \in \mathbb{R}^{d_a}$ is a bias term, and $\mathbf{w} \in \mathbb{R}^{d_a}$ is a learned vector projecting the hidden attention space to a scalar importance score.

- **Temporal Integration:** The final temporally-attended feature vector is computed as:

$$F_{\text{attended}} = \sum_{t=1}^{T} \alpha_t \cdot F_t \tag{13}$$

This operation integrates the temporal context into a single embedding vector, retaining the most salient motion cues that are crucial for fall detection.

3.4.3 Flattening and output

The resulting attended feature vector is passed through a *Flattening Layer*, producing a compact, 1D feature representation suitable for classification. This final output is referred to as **Feature Set V2** and can be used standalone or fused with complementary features (e.g., pose-based features) in a dual-stream setup.

As shown in Algorithm 2, the spatio-temporal feature learning leverages a 3D CNN backbone followed by a temporal attention mechanism to selectively emphasize key frames relevant for fall detection. The proposed attention-guided 3DCNN model achieves an effective trade-off between comprehensive spatiotemporal feature learning and computational focus on temporally informative frames. While the 3DCNN learns generic motion patterns, the attention mechanism allows the model to selectively focus on brief but critical periods, such as the moment of imbalance or impact during a fall.

Algorithm 2 Spatio-Temporal Feature Learning via 3D CNN with Temporal Attention

Input: Video sequence $X \in \mathbb{R}^{T \times H \times W \times C}$
Output: Temporally-attended feature vector F_{attended}
$Z^{[1]} \leftarrow \text{ReLU}(\text{Conv3D}_1(X))$ ▷ First 3D convolution
$A^{[1]} \leftarrow \text{MaxPooling}_1(Z^{[1]})$ ▷ First 3D max pooling
$Z^{[2]} \leftarrow \text{ReLU}(\text{Conv3D}_2(A^{[1]}))$ ▷ Second 3D convolution
$A^{[2]} \leftarrow \text{MaxPooling}_2(Z^{[2]})$ ▷ Second 3D max pooling
$A \leftarrow A^{[2]}$ ▷ Output feature map
for $t = 1$ to T **do**
  $F_t \leftarrow \text{Flatten}(A_t)$ ▷ Flatten each frame's feature map
$F \leftarrow \text{Stack}(F_1, F_2, \ldots, F_T)$ ▷ Shape: $T \times (HWD)$
for $t = 1$ to T **do**
  $e_t \leftarrow \mathbf{w}^\top \cdot \tanh(\mathbf{W}_{att} F_t + \mathbf{b}_{att})$ ▷ Attention score for frame t
$\alpha_t \leftarrow \text{softmax}(e_t)$ for all t ▷ Normalize attention weights
$F_{\text{attended}} \leftarrow \sum_{t=1}^{T} \alpha_t \cdot F_t$ ▷ Weighted sum of frame features
return F_{attended}

3.5 Attentional feature fusion

As illustrated in Fig. 1, the proposed system generates two complementary feature vectors: **Feature V1**, derived from handcrafted pose-based features (e.g., joint angles, distances, motion trajectories, acceleration), and **Feature V2**, obtained from deep spatiotemporal features extracted using a 3DCNN with temporal attention. These two representations capture different aspects of human activity-V1 emphasizes explicit

skeletal motion patterns, while V2 encapsulates visual and temporal dynamics from video frames.

To optimally leverage the strengths of both representations, an Attentional Feature Fusion mechanism is employed. This module dynamically learns to weight and integrate features from each stream, yielding a single, discriminative embedding suitable for classification.

3.5.1 Fusion formulation

Lets $\mathbf{v}_1 \in \mathbb{R}^{d_1}$ denote Feature V1 (pose-based handcrafted features) and $\mathbf{v}_2 \in \mathbb{R}^{d_2}$ denote Feature V2 (deep-learned features from 3DCNN with attention).

First, both feature vectors are projected into a common embedding space of dimension d_f using learnable linear transformations:

$$\tilde{\mathbf{v}}_1 = \mathbf{W}_1\mathbf{v}_1 + \mathbf{b}_1, \quad \tilde{\mathbf{v}}_2 = \mathbf{W}_2\mathbf{v}_2 + \mathbf{b}_2 \tag{14}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_f \times d_1}$, $\mathbf{W}_2 \in \mathbb{R}^{d_f \times d_2}$ are projection matrices, and $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^{d_f}$ are biases.

Next, attention weights are computed to determine the relative importance of each feature stream. The attention scores are obtained using a softmax-normalized scoring function:

$$\begin{aligned} \alpha_1 &= \frac{\exp(\mathbf{w}^\top \tanh(\tilde{\mathbf{v}}_1))}{\exp(\mathbf{w}^\top \tanh(\tilde{\mathbf{v}}_1)) + \exp(\mathbf{w}^\top \tanh(\tilde{\mathbf{v}}_2))}, \\ \alpha_2 &= 1 - \alpha_1 \end{aligned} \tag{15}$$

where $\mathbf{w} \in \mathbb{R}^{d_f}$ is a learnable parameter vector.

The final fused representation is computed as a convex combination of the two projected features:

$$\mathbf{v}_{\text{fused}} = \alpha_1 \cdot \tilde{\mathbf{v}}_1 + \alpha_2 \cdot \tilde{\mathbf{v}}_2 \tag{16}$$

This fused feature vector $\mathbf{v}_{\text{fused}} \in \mathbb{R}^{d_f}$ is then passed through subsequent classification stage.

The same process of *Attentional Feature Fusion* has been detailed in Algorithm 3. As shown in Algorithm 3, the proposed fusion mechanism adaptively combines handcrafted pose features and deep spatiotemporal features into a unified embedding. The attention weights α_1 and α_2 are dynamically learned, enabling the model to emphasize the most informative representation for each instance. This fusion strategy enhances the discriminative power by jointly capturing interpretable skeletal motion cues and rich visual-temporal dynamics.

3.6 Classification and output

The fused feature vector $\mathbf{x} \in \mathbb{R}^n$ is passed through two fully connected dense layers. The first dense layer applies a linear transformation followed by a ReLU activation:

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) \tag{17}$$

where $\mathbf{W}_1 \in \mathbb{R}^{m \times n}$ and $\mathbf{b}_1 \in \mathbb{R}^m$ are the weight matrix and bias vector, respectively. Dropout regularization with a rate of 30% is applied to $\mathbf{h}_1$ during training to prevent overfitting by randomly setting 30% of its elements to zero.

The resulting vector is then fed into the second dense layer:

$$\mathbf{h}_2 = \text{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2) \tag{18}$$

with $\mathbf{W}_2 \in \mathbb{R}^{k \times m}$ and $\mathbf{b}_2 \in \mathbb{R}^k$.

Finally, a Softmax layer produces the probability distribution over the two classes-*Fall* and *No Fall*:

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3) \tag{19}$$

where

$$\text{Softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad i \in \{1, 2\} \tag{20}$$

and $\mathbf{W}_3 \in \mathbb{R}^{2 \times k}$, $\mathbf{b}_3 \in \mathbb{R}^2$.

When an individual's fall is detected, an alert notification is triggered for caregivers or monitoring personnel alongside annotated video clips capturing determination highlights for real-time video captioning interpretability. In case there is no fall, the system remains in a non-invasive monitoring mode with no alerts being raised.

Algorithm 3 Attentional Feature Fusion

Input: Pose-based feature vector $\mathbf{v}_1 \in \mathbb{R}^{d_1}$, Deep feature vector $\mathbf{v}_2 \in \mathbb{R}^{d_2}$

Output: Fused feature vector $\mathbf{v}_{\text{fused}} \in \mathbb{R}^{d_f}$

Project features into common space:
$\tilde{\mathbf{v}}_1 \leftarrow \mathbf{W}_1\mathbf{v}_1 + \mathbf{b}_1$
$\tilde{\mathbf{v}}_2 \leftarrow \mathbf{W}_2\mathbf{v}_2 + \mathbf{b}_2$
Compute attention weights:
$\alpha_1 \leftarrow \text{softmax}(\mathbf{w}^\top \tanh(\tilde{\mathbf{v}}_1))$
$\alpha_2 \leftarrow 1 - \alpha_1$
Fuse features:
$\mathbf{v}_{\text{fused}} \leftarrow \alpha_1 \cdot \tilde{\mathbf{v}}_1 + \alpha_2 \cdot \tilde{\mathbf{v}}_2$
return $\mathbf{v}_{\text{fused}}$

4 Results and discussion

4.1 Dataset

KFall dataset [4] greatly aids advanced studies focused on fall detection and injury prevention among the elderly. The dataset allows information to be obtained from 32 subjects who undertook 21 distinct activities of daily living (ADL)

Table 1 Non-falls and falls scenarios

S. No.	Non-falls	Falls
1	Stand	Forward fall when trying to sit down
2	Stand, slowly bend the back with or without bending at the knees, tie shoelaces, and get up	Backward fall when trying to sit down
3	Gently jump	Lateral fall when trying to sit down
4	Stand, sit on the ground, wait a moment, and get up at normal speed	Forward fall when trying to get up
5	Walk normally with a turn	Lateral fall when trying to get up
6	Walk quickly	Forward fall while sitting, caused by fainting
7	Jogging normally	Lateral fall while sitting, caused by fainting
8	Jogging quickly	Backward fall while sitting, caused by fainting
9	Stumble while walking	Vertical (forward) fall while walking, caused by fainting
10	Sit on the chair for 30 s	Fall while walking, use of hands to dampen fall, caused by fainting
11	Sit on the sofa (back is inclined to the support) for 30 s	Forward fall while walking caused by a trip
12	Sit down in a chair normally and get up from a chair normally	Forward fall while jogging caused by a trip
13	Sit down to a chair quickly, and get up from a chair quickly	Forward fall while walking caused by a slip
14	Sit a moment, trying to get up, and collapse into a chair	Lateral fall while walking caused by a slip
15	Stand, sit on the sofa (back is inclined to the support), and get up normally	Backward fall while walking caused by a slip
16	Lie on the bed for 30 s	
17	Sit a moment, lie down on the bed normally, and get up normally	
18	Sit a moment, lie down on the bed quickly, and get up quickly	
19	Walk upstairs and downstairs normally (5 steps)	
20	Walk upstairs and downstairs quickly (5 steps)	

alongside 15 fall scenarios. These activities and conditions are quintessential to the human movement capture for detecting falls and the subsequent research, as discussed in Table 1. The dataset has 5075 records of movements, 2729 of which are ADL movements, while 2346 are fall movements. Hence, researchers can design and test advanced algorithms for fall detection that must distinguish between ADLs and falls. The KFall dataset is arguably distinguished for its synchronized video references that depict labeled falls for each action video.

These labels enable improved usability and accuracy of the dataset, improving the accuracy of developing and accessing algorithms by characterizing each fall's exact onset and impact information. KFall allows for the seamless integration of movement data related to actual falls captured in video footage with the movement being filmed. This helps develop advanced algorithms that can detect and classify falls efficiently and accurately in various situations. The dataset is split into two segments: an 80% portion is allocated for training the model, while a 20% portion is reserved for testing the model. The total number of records within the test subset amounted to 1015, with 546 corresponding to non-fall (ADL) movements and 469 corresponding to fall movements. The KFall dataset is highly beneficial for developing systems that efficiently detect falls and support injury prevention among elderly individuals. Its comprehensive resources, including diverse fall scenarios and detailed motion data, significantly enhance the quality and effectiveness of research and development efforts. This makes the dataset an essential asset in advancing healthcare solutions and assistive technologies to improve the ageing population's safety and well-being.

Table 1 outlines a comprehensive set of scenarios categorized into non-fall and fall activities. The non-fall cases include everyday movements such as standing, walking, sitting, and lying down. In contrast, the fall scenarios simulate

various accidental events, including slips, trips, and fainting-induced falls across multiple postures.

4.2 Model hyperparameter tuning

With regards to performance, model hyperparameter tuning is important, especially when considering factors such as training speed and prediction accuracy. In this case, the dropout rate is selected to be 0.3, which helps prevent overfitting by randomly deactivating hidden layer neurons during the training. Training is performed in 100 epochs to ensure sufficient learning while avoiding overtraining, adjusting the batch size to 16 to balance the computational burden and steady parameter update. Overfitting is mitigated with a dropout rate of 0.3 on the neuron level. Non-linearity and additional convergence acceleration of the ReLU (Rectified Linear Unit) activation function are introduced. At the same time, convolutional layers extract spatial features efficiently using a kernel and pooling layer sized (3,3,3) and (2,2,2), respectively. The latter decreases the dimensions of the feature map while retaining crucial information. Training effectiveness and stability are improved by employing the Adam optimizer, where the adaptive learning rate makes it highly sought after. Gamma contrast augmentation is applied to enhance generalisation with a range from 0.5 to 2.0. The minimum detection and tracking confidence thresholds are set at 0. 6 to filter out low-confidence predictions. These hyperparameters are adapted to provide a satisfactory balance between immediate processing and smooth visual movement at the 15fps cap for output video frame rate, leading to increased effectiveness during inference processes.

All the experiments are done using a Windows 11 machine with the following specs: 16 GB of RAM, a 512 GB SSD, and an NVIDIA RTX 4050 6 GB graphics card. TensorFlow and Keras were used for deep learning in Python 3 and other supported libraries to develop and train the model.

4.3 Result analysis

To test the performance of the fall detection system guided by the dual stream attention framework, we performed extensive tests using the KFall benchmark dataset. The model's performance is evaluated using the standard evaluation metrics of accuracy, precision, and recall to identify both fall and non-fall events. These measures give insight into the functionality of a system in discerning critical from noncritical activities.

As mentioned in Table 2, the system has acquired an astonishing accuracy of 98.71%, taking all classes into consideration. The model is 99% accurate (precision) about all predictions made for falling. Also, the model missed 98% as the correct fall (recall). Regarding the class of non-falling, the precision is 98%, and the recall is 0.99%. The F1-score

Table 2 Performance metrics for fall and non-fall classes

Metric	Fall	Non-fall
Precision	0.99	0.98
Recall	0.98	0.99
F1-Score	0.98	0.98
Accuracy	98.71%	

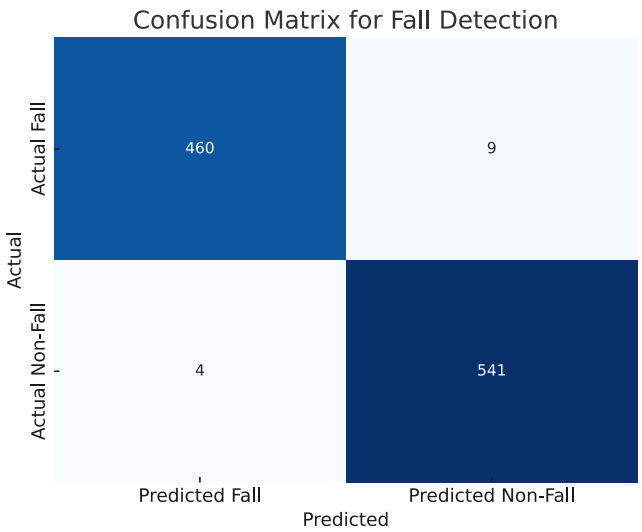


Fig. 4 Confusion matrix of the 3D CNN with temporal attention

of 0.98 for both classes proves that the model is trustworthy when used in practical applications.

The deep analysis of the model's predictive distribution, including true positives, true negatives, false positives, and false negatives, is derived from the confusion matrix in Fig. 4. The model is able to correctly detect 460 fall instances and 541 non-fall instances, while erroneously predicting nine instances as false positives and four as false negatives. The model's low error rates showcase its ability to avoid critical misclassifications, an important aspect of safety-centric functions such as elder monitoring.

Both training and validation accuracy trends have been plotted in Fig. 5. It can be observed that both curves have a steady convergence with a small gap in between, which is indicative of good model generalization and lack of overfitting. This situation occurs because of careful architecture, data handling, and dropout use.

To prove how the model would work in real life, we tested cases in videos where a person fainted. Figure 6 shows how the fall is captured by the model, and we are alerted immediately via a monitor, as it should have been. Figure 7, on the other hand, demonstrates the system's ability to accurately label walking movements as non-falls without any external alerts. The findings prove that the framework works excellently and reliably under many varying conditions.

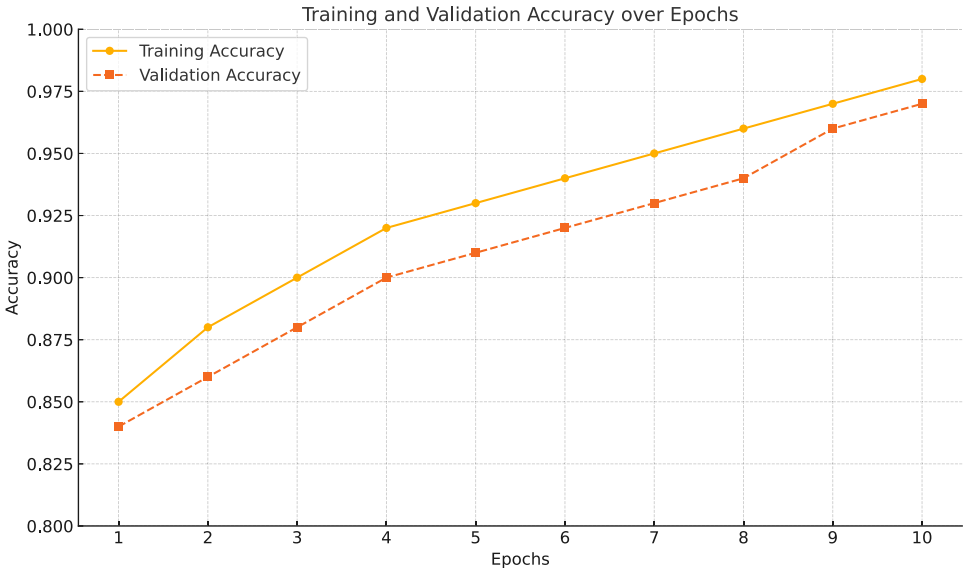


Fig. 5 Training and validation accuracy over epochs



Fig. 6 Detected fall event during fainting scenario

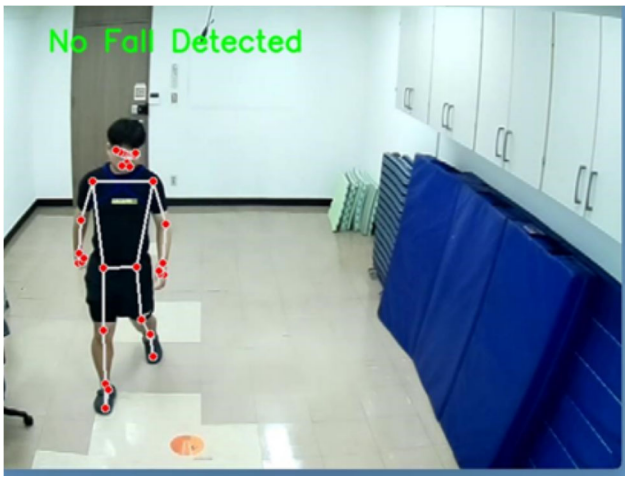


Fig. 7 Correct classification of non-fall activity (walking)

In summary, both quantitative and qualitative evidence support the effectiveness of the proposed dual-stream framework. Integrating pose-based motion signals and deep spatial-time features with temporal attention enables the system to recognize falls remarkably and sensitively. This makes it a strong candidate for real-world deployment in intelligent healthcare monitoring and assistive monitoring environments.

4.4 Ablation study

The effectiveness of the proposed fall detection framework is evaluated by comparing it with two baseline architectures, as shown in Fig. 8. The proposed model leverages a dual-stream architecture, which includes both skeletal pose data and raw video frames, and achieved the highest accuracy (98.71%), precision (98.08%), and recall (99.13%). When evaluated without the pose stream (3DCNN + FFNN), the accuracy dropped significantly to 93.5%, highlighting a 5.21% performance loss due to the absence of posture information. Similarly, using only the combined features with FFNN resulted in an accuracy of 95.8%, showing a 2.91% drop from the full model. These results confirm the advantage of jointly learning both spatial and skeletal dynamics for discriminative fall detection in complex indoor settings.

The training configuration further affects the effectiveness of the model, particularly the choice of optimizer and batch size. Figure 9 shows a radar (spider) plot comparing the performance of four optimizers: Adam, SGD, RMSprop and AdaDelta at batch sizes of 4, 8, 16 and 32. Among them, the Adam optimizer achieved the best overall performance, reaching a peak accuracy of 98.71% at batch size 16. RMSprop followed closely, although with slightly less stability across batch sizes. In contrast, SGD showed moderate

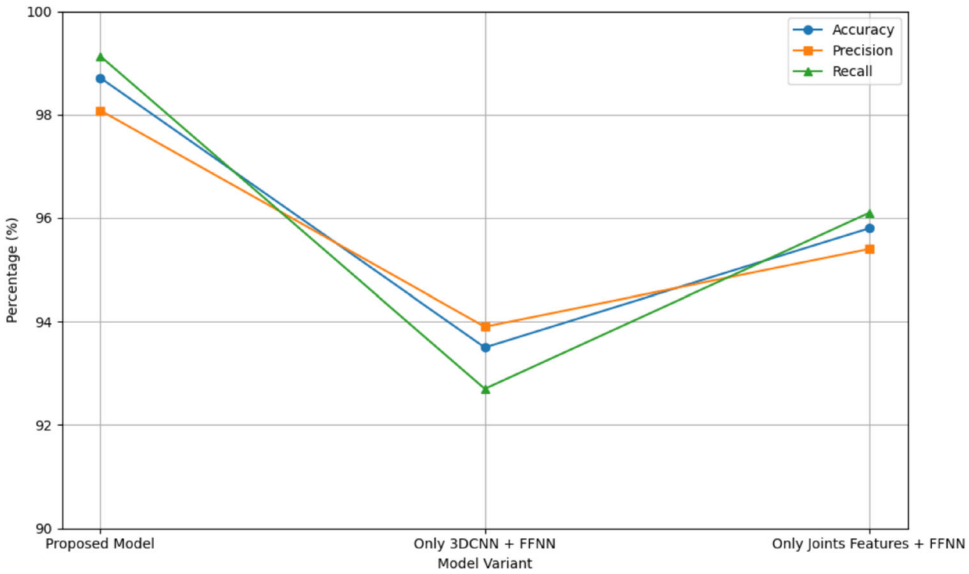


Fig. 8 Comparative performance of model variants using key classification metrics

improvement but lacked stability, while AdaDelta's behavior is inconsistent and experienced a sharp drop in accuracy of more than 8% between its best and worst performing batch sizes.

Overall, these experimental results demonstrate the strong synergy that the proposed dual-stream design enabled and emphasize the importance of careful training configuration. The observed accuracy drops in simplified variants confirm that fall detection tasks benefit significantly from multimodal feature fusion, especially when temporal dynamics and skeletal joint movements are jointly modeled. Furthermore, optimizer selection and batch size tuning are critical in maximizing performance and ensuring model generalization across diverse input conditions.

4.5 Comparison with existing methods

Recent human activity recognition (HAR) advancements have explored diverse architectures to improve classification accuracy and robustness. Ansari and Singh [13] demonstrated that optimized hyperparameter tuning in recurrent neural networks significantly enhances model convergence and classification accuracy, achieving around 94.6% accuracy. Kushwaha and Khare [16] combined optical flow with appearance-based features to capture spatio-temporal dynamics more effectively, reporting an accuracy near 92.3%. By applying URILBP and ConvSTLSTM to a multiview framework, Kurchaniya and Kumar [12] suggested appearance and motion-based methods, which achieved almost 96.2% accuracy. Ansari et al. [3] developed PosturalNet, which incorporates detailed postural cues, reaching an accuracy of 95.4%.

The proposed dual-stream attention-guided framework advances this work by integrating skeletal pose estimation with raw video inputs through a 3D CNN enhanced with temporal attention mechanisms. This fusion enables the model to selectively emphasize discriminative motion segments and posture features, resulting in a significant performance boost to 98.71% accuracy on benchmark datasets. Figure 10 illustrates this performance gain, confirming that combining complementary spatial and temporal features through attention mechanisms can substantially enhance fall detection reliability compared to prior methods.

5 Conclusion and future direction

This paper presents a novel dual-stream attention-guided framework for indoor fall detection that effectively combines skeletal pose information with raw video features. Incorporating these complementary data streams allows the model to capture the intricate postural dynamics and rich spatio-temporal motion patterns, thus enabling greater accuracy and robustness in fall recognition. Adding a temporal attention mechanism narrows the model's focus even further to the important action portions, thus improving interpretability and performance. Results from the experiments performed on the KFall dataset validate the superiority of the proposed method, achieving 98.71% accuracy. Plus, the real-time alert system highlights the usefulness of this framework in smart homes, elder care, and even general health monitoring. The dual-stream architecture thus represents a significant advancement by bridging the gap between pose-based and video-based fall detection techniques.

This model has sensitivity to video quality, where low resolution or noisy inputs can degrade pose estimation accuracy and overall performance. Several directions can be explored for future work to improve the system further. Expanding

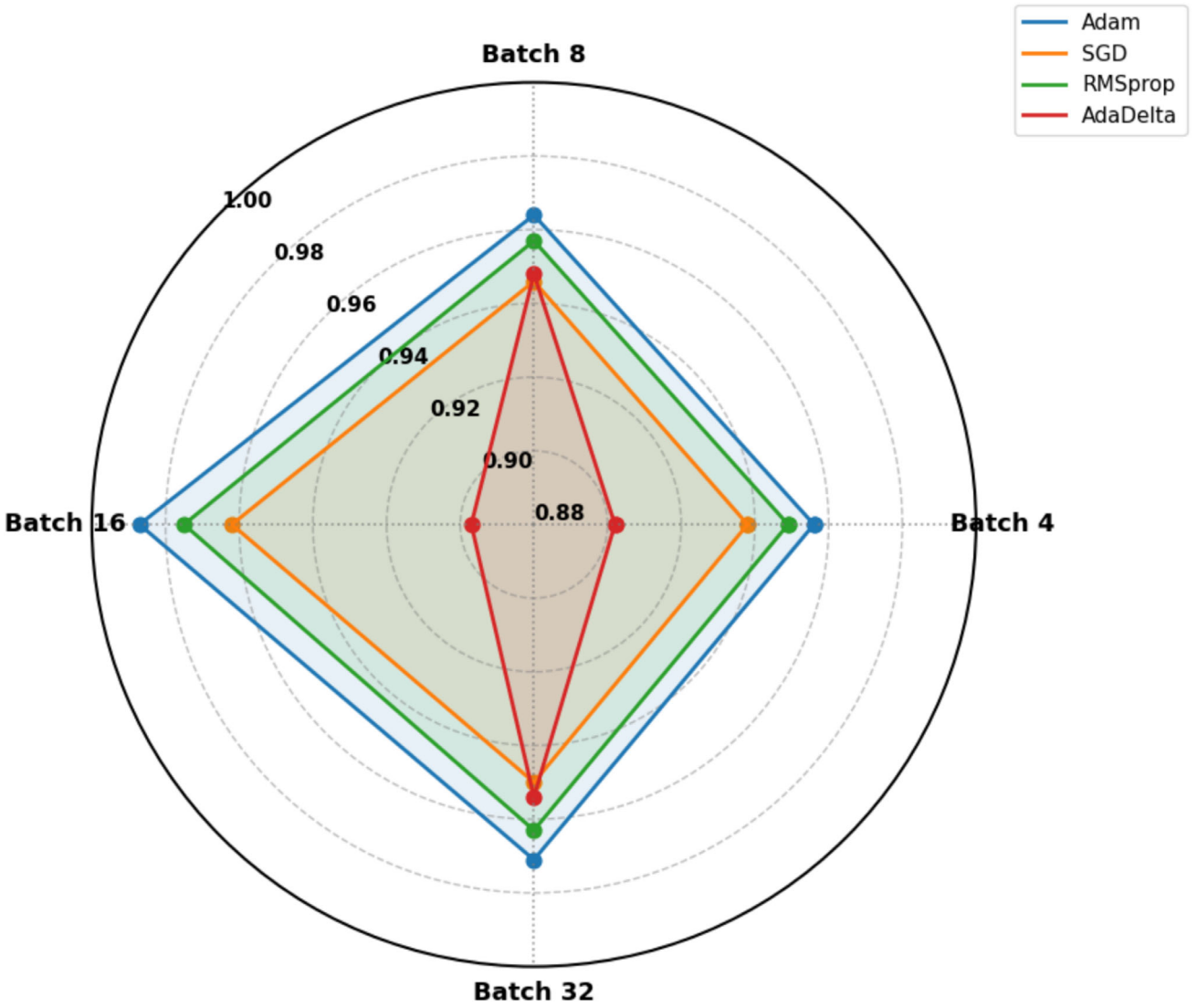


Fig. 9 Radar plot of optimizer performance across varying batch sizes

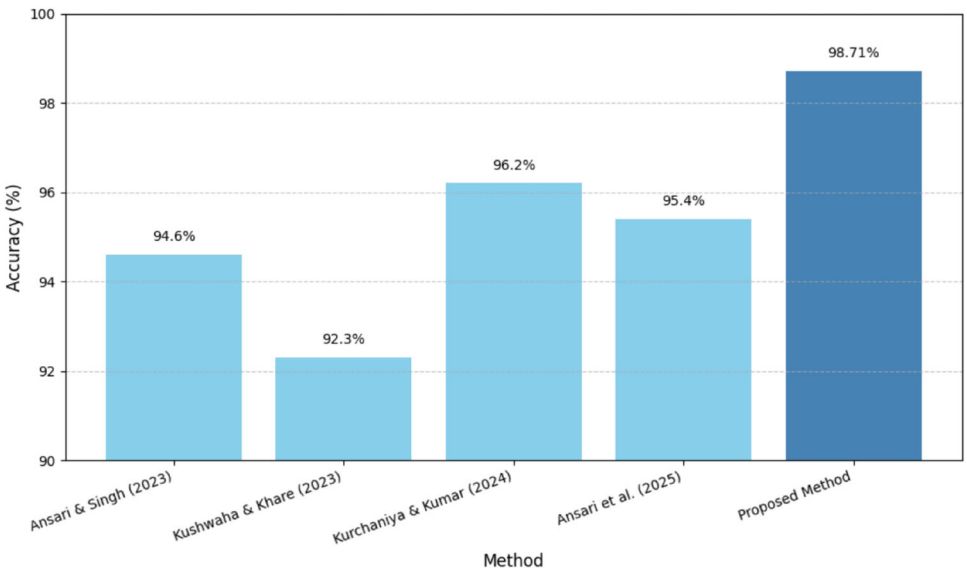


Fig. 10 Comparative analysis among existing methods

the dataset with more diverse fall scenarios and environmental conditions would enhance generalizability. MediaPipe's performance may degrade under low-light or occluded conditions due to its reliance on RGB input. Incorporating multiview or multi-camera setups could address occlusion and improve spatial coverage. Additionally, integrating lightweight model optimization techniques will facilitate deployment on edge devices with limited computational resources. Finally, extending the framework to recognize a broader range of abnormal activities beyond falls could increase its utility in comprehensive health monitoring systems. Apart from that, future research has sufficient scope to focus on integrating multimodal data such as audio signals and inertial sensor measurements to enhance the accuracy and robustness of activity recognition systems. Advanced architectures like transformers offer significant potential for this integration, as their self-attention mechanisms can effectively capture complex temporal and spatial dependencies across modalities. Exploring multimodal transformer models, hybrid CNN-transformer designs, and lightweight transformer variants will enable more powerful and efficient feature fusion, ultimately improving system performance and enabling real-time deployment in diverse applications.

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Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this paper.

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